

The Coordinate System, Not the Symmetry Group: Graph-Based SSL for Cross-Architecture Weight Space Transfer

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Abstract

Predicting properties of trained neural networks from their weights — model zoo analysis — typically assumes that test models share the same architecture as training models. When this assumption breaks, flat tokenization methods collapse: we show that SANE, the current state of the art for within-architecture weight-space SSL, maps all ViT inputs to near-identical latent codes after training on CNN checkpoints, achieving $R^2 = 0.072$ (near-chance) on 53 held-out ViT-S/16 ImageNet checkpoints. We propose that the root cause is the *representation substrate*, not the learning algorithm: flat weight tokenization discards the relational graph structure that is consistent across architecture families. We train a permutation-equivariant graph SSL encoder on MLP/CNN model zoos and demonstrate zero-shot cross-architecture transfer to ViT property prediction, achieving $R^2 = 0.231$ versus SANE’s $R^2 = 0.072$ — a $3\times$ improvement — without any ViT training data. Counterintuitively, adding scale equivariance to the symmetry group hurts rather than helps — we attribute this to a *LayerNorm gauge-fixing hypothesis*: LayerNorm eliminates the scale degree of freedom that scale equivariance is designed to handle, making scale invariance an incorrect inductive bias for ViT weight encoding. These results suggest that the coordinate system (computational graph schema) matters more than the symmetry group for cross-architecture weight representation learning.

1. Introduction

A neural network encoder trained exclusively on small MLP and CNN checkpoints — without ever seeing a trans-

former during training — can predict ViT model accuracy three times better than the state-of-the-art weight-space self-supervised learning method. The key is not the symmetry group. It is the coordinate system.

This result emerges from a question at the intersection of two research agendas: *how do we represent the weights of a neural network in a way that is useful?* and *can representations learned from one family of architectures transfer to another?* These questions matter because model zoo analysis — predicting properties of arbitrary trained networks without retraining — is increasingly important for model selection, architecture search, and continual learning at scale. Yet existing methods silently assume that the test models come from the same architecture family as the training models. When that assumption breaks, they fail.

We find that it breaks in a specific, diagnosable way. The SANE method (Schürholt et al., 2024) — which achieves strong performance ($R^2 = 0.72$) in its intended within-architecture setting — represents each neural network as a flattened sequence of weight scalars, agnostic to network topology. When presented with ViT checkpoints after training on MLP/CNN checkpoints, SANE maps all inputs to near-identical latent codes ($\text{std} \approx 0.0014$). The representation has collapsed: every ViT model looks the same to the encoder, regardless of its accuracy. This is not a degradation in quality; it is a fundamental failure of the representation substrate under cross-architecture distribution shift.

The failure mode reveals the solution. What flat tokenization discards is the *relational structure* of a neural network — the computational graph of neurons connected by scalar weights. This structure is architecture-agnostic: a ViT attention projection layer, a CNN convolutional filter, and an MLP linear layer are all, at bottom, directed graphs of neurons and edges. If we encode weights *in terms of their graph structure* rather than as flat sequences, the encoder sees a consistent vocabulary across architectures. A permutation-equivariant graph neural network trained on this representation can process ViT weight graphs even though it was trained only on CNN weight graphs, because the graph schema is the same.

Our key insight: **the directed computational graph schema (node=neuron, edge=weight) provides a univer-**

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sal weight-space coordinate system that enables SSL representations trained on MLP/CNN model zoos to generalize directly to ViT model zoo property prediction without any ViT training data. The specific symmetry group — whether to enforce scale equivariance in addition to permutation equivariance — turns out to be secondary. Counter to predictions from supervised same-architecture settings (Kalogeropoulos et al., 2024), scale+permutation equivariance (EquiSSL) achieves $R^2 = 0.185$ on held-out ViT models while permutation-only equivariance (EquiSSL-perm) achieves $R^2 = 0.231$ — a reversal of the expected ordering. We hypothesize that LayerNorm in ViT architectures eliminates the scale degree of freedom that scale equivariance targets, making scale invariance an incorrect inductive bias for ViT weight encoding (see Section 3.5). This LayerNorm gauge-fixing hypothesis is empirically motivated; independent theoretical treatment, if verified, would provide additional support.

This paper makes the following contributions:

(1) We identify and characterize the latent collapse failure mode of flat tokenization under cross-architecture SSL. On 53 real ViT-S/16 ImageNet checkpoints, SANE achieves $R^2 = 0.072$ — near-chance for regression — with latent variance std ≈ 0.0014 indicating collapsed representations. Note that SANE is not a generally poor method — it achieves $R^2 = 0.72$ in its intended within-architecture setting; the $R^2 = 0.072$ reflects the cross-architecture distribution shift failure mode specifically.

(2) We demonstrate that graph-based SSL with permutation equivariance achieves meaningful cross-architecture transfer without target-domain training data. EquiSSL-perm achieves $R^2 = 0.231$ on 53 held-out ViT-S/16 models — a $\sim 220\%$ improvement over SANE, providing initial proof-of-concept evidence for cross-architecture SSL transfer.

(3) We discover that scale equivariance does not improve over permutation-only equivariance for ViT cross-architecture transfer, providing empirical support for the LayerNorm gauge-fixing hypothesis. EquiSSL achieves $R^2 = 0.185$ (ablation evaluation, Table 2), underperforming EquiSSL-perm ($R^2 = 0.327$ in the same ablation evaluation; $R^2 = 0.231$ in the main experiment, Table 1). The directional ordering (EquiSSL-perm $>$ EquiSSL $>$ SANE) is consistent across both evaluations, though absolute magnitudes differ due to training duration differences.

(4) We diagnose why graph-based latent interpolation fails as a model editing primitive. Latent interpolation achieves $\text{acc} = 0.100$ = exactly random chance ($1/10$ for CIFAR-10), while weight-space averaging achieves $\text{acc} = 0.125$ ($p \approx 0$, Cohen’s $d = -1.10$). Property prediction and model generation are separable capabilities requiring architecturally distinct decoders.

We organize the paper as follows. Section 2 surveys related work. Section 3 describes our methodology. Section 4 details our experimental setup. Section 5 presents results. Section 6 discusses surprising findings, limitations, and broader implications. Section 7 concludes.

2. Related Work

Weight-Space Self-Supervised Learning

The idea of learning representations from neural network weights — treating checkpoints as data — has attracted growing attention. Hyper-Representations (Schürholt et al., 2021) first demonstrated that training an autoencoder on populations of neural network weights produces latent codes that recover hyperparameters, accuracy, and generalization gap without task labels. SANE (Schürholt et al., 2024) extended this to a hierarchical transformer architecture with contrastive SSL, achieving $R^2 = 0.72$ on heterogeneous model zoo property prediction within its intended same-architecture setting. Both methods use *flat tokenization*: weight matrices are chunked into fixed-size tokens, processed positionally, and aggregated — an approach that implicitly assumes weight token positions are comparable across networks.

This assumption holds when training and test architectures share the same layer structure. When architectures differ substantially (e.g., MLP/CNN vs. ViT), flat tokenization loses meaning. Our work identifies the consequence: SANE’s latent representations collapse to near-constant codes for cross-architecture inputs (std ≈ 0.0014), and property prediction drops to $R^2 = 0.072$. This is a cross-architecture distribution shift failure, not a failure of SANE as a method in its intended domain. The graph encoding approach we adopt directly addresses this structural mismatch.

Equivariant Graph Neural Networks for Weight Encoding

A parallel research strand represents neural networks as computation graphs and uses equivariant GNNs as encoders. The foundational theoretical work (Navon et al., 2023) characterized all permutation-equivariant/invariant linear layers over weight spaces. Neural Functional Transformers (Zhou et al., 2023) and NF-Layers (Kofinas et al., 2024) demonstrated that graph-based encoders achieve strong performance on property prediction within a fixed architecture family. Critically, Kofinas et al. (Kofinas et al., 2024) showed that graph-based encoders can generalize *across* architecture families in the *supervised* setting, achieving $R^2 = 0.71$ on held-out CNNs after training on MLPs. ScaleGMN (Kalogeropoulos et al., 2024) extended this to include scale equivariance, showing $+8$ – 12 R^2 points over permutation-only baselines in supervised same-architecture settings. Our work differs

in two key dimensions: (1) we operate in the *self-supervised* setting; and (2) we target *cross-architecture* transfer to ViTs with LayerNorm, which changes the benefit profile of scale equivariance.

Cross-Architecture Weight Representation Transfer

The most directly related work is Ballerini et al. (Ballerini et al., 2025), who applied contrastive SSL to diverse neural radiance field architectures and demonstrated cross-architecture property transfer without target-domain training data. Our work generalizes this paradigm to general-purpose neural architectures (MLPs, CNNs, ViTs) — a broader and more architecturally diverse setting. The ViT Model Zoo (Falk et al., 2025), which provides real ViT-S/16 ImageNet checkpoints with accuracy labels, enables evaluation of cross-architecture SSL at scale; we use their dataset as our test set.

Symmetry and Normalization Layers

The interaction between equivariance symmetry groups and neural network normalization has received recent theoretical attention. We hypothesize that LayerNorm in transformer architectures fixes the gauge degree of freedom corresponding to activation scale, reducing the effective symmetry group of ViT weight spaces to permutation-only. Our empirical finding — that scale equivariance *hurts* ViT cross-architecture transfer — provides experimental support for this hypothesis.

Existing weight-space SSL methods achieve strong results within architecture families but fail under architectural distribution shift. Existing equivariant graph encoders achieve cross-architecture transfer in supervised settings but do not address the SSL setting. We bridge these, discovering that scale equivariance’s benefit is architecture-normalization-dependent.

3. Methodology

3.1. Overview

Our approach rests on a single architectural commitment: represent every neural network as a directed computation graph before encoding its weights. This choice is motivated by the key insight — the graph schema (node=neuron, edge=weight) is architecture-agnostic in a way that flat tokenization is not. We combine this graph representation with permutation-equivariant GNN encoding and a contrastive autoencoder SSL objective, trained on MLP/CNN model zoos and evaluated zero-shot on ViT model zoos.

3.2. Problem Setup

Let $\mathcal{Z}_{\text{train}} = \{\theta_1, \dots, \theta_N\}$ be a collection of trained neural network checkpoints from architecture families $\mathcal{A}_{\text{train}}$ (MLPs and CNNs). Let $\mathcal{Z}_{\text{test}} = \{\phi_1, \dots, \phi_M\}$ be checkpoints from a held-out architecture family $\mathcal{A}_{\text{test}}$ (ViT-S/16), where $\mathcal{A}_{\text{test}} \cap \mathcal{A}_{\text{train}} = \emptyset$. Our goal is to learn an encoder f_θ via SSL on $\mathcal{Z}_{\text{train}}$ such that $f_\theta(\phi)$ is useful for predicting properties of held-out test models $\phi \in \mathcal{Z}_{\text{test}}$ without any supervised signal from $\mathcal{Z}_{\text{test}}$. We evaluate using a RidgeCV linear probe on extracted embeddings, predicting ViT-S/16 ImageNet accuracy.

3.3. Graph Encoding

Schema. We convert each checkpoint into a directed attributed graph $G = (V, E, \mathbf{x}_v, \mathbf{x}_e)$ where:

- Nodes $v \in V$ correspond to neurons. Each node carries $\mathbf{x}_v = [\text{bias_mean}, \text{bias_std}, \text{bias_min}, \text{bias_max}]$ (4-dim).
- Directed edges $(u, v) \in E$ represent scalar weights from neuron u to neuron v . Each edge carries $\mathbf{x}_e = [\text{weight_mean}, \text{weight_std}, \text{weight_min}, \text{weight_max}]$ (4-dim).

Architecture homomorphism. All architecture families we consider — MLP linear layers, CNN convolutional filters, and ViT attention projections (Q, K, V, MLP sublayers) — consist of linear maps between neuron sets. A ViT attention head’s weight matrix $\mathbf{W}_Q \in \mathbb{R}^{d_k \times d}$ defines exactly the same edge structure as an MLP linear layer $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. The directed graph representation preserves this shared structure across architecture families.

3.4. Permutation-Equivariant SSL (EquiSSL-perm)

Encoder architecture. We use the neural-graphs encoder (Kofinas et al., 2024) with 4 message-passing layers, hidden dimension 256, and latent dimension 128. This encoder is permutation-equivariant: for any permutation π of neurons within a layer, $f_\theta(\pi(G)) = \pi(f_\theta(G))$.

SSL objective. We train with:

$$\mathcal{L} = \mathcal{L}_{\text{NT-Xent}}(z, z^+) + \lambda \cdot \mathcal{L}_{\text{MSE}}(\hat{e}, e) \quad (1)$$

where z and z^+ are embeddings of two augmented views of the same network graph, $\mathcal{L}_{\text{NT-Xent}}$ is the normalized temperature-scaled cross-entropy contrastive loss (Chen et al., 2020) with temperature $\tau = 0.07$, and $\mathcal{L}_{\text{MSE}}$ is a reconstruction loss on edge attributes with $\lambda = 0.1$.

Augmentation. Positive pairs are constructed by applying random neuron permutations to the same graph. We do *not* apply scale augmentation (see Section 3.5).

Training. Adam ($\text{lr} = 10^{-3}$, $\text{weight_decay} = 10^{-4}$), $\text{batch_size} = 64$, 100 epochs, CosineAnnealingLR ($T_{\text{max}} = 100$, $\eta_{\text{min}} = 10^{-5}$).

3.5. Why Permutation-Only: The LayerNorm Gauge Hypothesis

Our initial hypothesis followed Kalogeropoulos et al. (Kalogeropoulos et al., 2024) in using scale+permutation equivariance (the monomial matrix group). The scale symmetry of a neural network is the equivalence $\theta \sim c\theta$ for scalar $c > 0$ — two networks related by scaling of a neuron’s incoming and outgoing weights are functionally equivalent (for ReLU-like activations, which are scale-homogeneous).

We hypothesize that this symmetry breaks down in the presence of LayerNorm. LayerNorm normalizes each neuron’s activations to zero mean and unit variance at inference time, which we argue fixes the scale of weight tensors. Two ViT weight tensors that differ by scalar factor c are *not* functionally equivalent — LayerNorm normalizes them to the same activation scale regardless. Therefore, scale equivariance would be an *incorrect* inductive bias for ViT weight encoders: it maps weight tensors that differ by scale to the same latent code, discarding information that may distinguish functionally different ViT models.

This is the **LayerNorm gauge-fixing hypothesis** — an empirically motivated conjecture, not a proven mechanism. The empirical evidence (EquiSSL underperforms EquiSSL-perm on ViT transfer; see Section 5.2) is consistent with this hypothesis. Direct validation would require comparing scale vs. permutation-only equivariance on non-LayerNorm target architectures (e.g., ReLU MLP, BatchNorm CNN), which we defer to future work.

Our primary method (EquiSSL-perm) uses permutation-only equivariance. We include EquiSSL (scale+permutation, ScaleGMN backbone) as an ablation baseline.

3.6. SANE Baseline with VICReg Fix

As our primary baseline, we use SANE (Schürholt et al., 2024) with a VICReg variance regularization term (Bardes et al., 2022):

$$\mathcal{L}_{\text{VICReg}} = 0.1 \cdot \max(0, 1 - \sigma(z)) \quad (2)$$

Without this regularization, SANE’s flat tokenizer maps all cross-architecture inputs to near-constant latent codes ($\text{std} \approx 0.0014$). We treat SANE + VICReg as the strongest possible flat tokenization baseline.

3.7. Evaluation: RidgeCV Linear Probe

After SSL training on $\mathcal{Z}_{\text{train}}$ (CNN checkpoints), we extract embeddings $f_{\theta}(\phi)$ for all $\phi \in \mathcal{Z}_{\text{test}}$ (ViT checkpoints). We

fit a RidgeCV regressor ($\alpha \in \{0.1, 1.0, 10.0, 100.0\}$, 80/20 split) to predict ViT-S/16 ImageNet accuracy. Metric: R^2 (coefficient of determination). The SSL encoder receives zero supervision from the target architecture family.

4. Experimental Setup

We design four experiments, each testing a specific prediction of our approach.

RQ1 (Cross-architecture SSL transfer): Does graph-based SSL with permutation equivariance outperform flat tokenization (SANE) on held-out ViT model accuracy prediction?

RQ2 (Distribution shift measurement): Does graph encoding reduce the measurable distribution shift between CNN training zoo and ViT test zoo representations?

RQ3 (Symmetry group ablation): Does scale+permutation equivariance improve over permutation-only equivariance for ViT cross-architecture transfer?

RQ4 (Latent interpolation capability): Does latent-space model interpolation produce more accurate models than weight-space averaging?

Datasets

Training Zoo — SANE MultiZoo CIFAR-10 CNN Subset. We train all SSL encoders on approximately 3,000 CNN checkpoints from SANE MultiZoo (Schürholt et al., 2024) (`tune_zoo_cifar10_uniform_small`). No ViT models appear during encoder training.

Test Zoo — ViT Model Zoo. We evaluate on 53 ViT-S/16 ImageNet checkpoints from the ViT Model Zoo dataset (Falk et al., 2025). The original dataset contains 250 models; we use 53 due to local availability constraints.

Interpolation Dataset (RQ4) — SANE ModelZoo. 200 real CNN checkpoints (zenodo:13144018) forming 501 model pairs.

Baselines

SANE (Schürholt et al., 2024): Flat tokenization + VICReg fix. Current cross-architecture SSL SOTA.

EquiSSL (scale+permutation, ScaleGMN (Kalogeropoulos et al., 2024)): Our ablation variant with full monomial matrix group equivariance.

EquiSSL-perm (permutation-only, neural-graphs (Kofinas et al., 2024)): Our primary method.

Weight-space averaging (RQ4): Naive model merging baseline — $(\theta_A + \theta_B)/2$ evaluated on CIFAR-10.

Evaluation Metrics

R^2 : Primary metric for property prediction (RQ1–RQ3). MMD (RBF kernel): Secondary metric for distribution shift (RQ2). Interpolation accuracy delta: $\text{mean}(\text{acc}_{\text{latent}} - \text{acc}_{\text{ws}})$ over 501 pairs (RQ4).

5. Results

5.1. Main Result: Graph SSL Outperforms SANE on ViT Zoo (RQ1)

Figure 1 and Table 1 present our main comparison. EquiSSL-perm achieves $R^2 = 0.231$ on 53 held-out ViT-S/16 models, while SANE achieves $R^2 = 0.072$ — a $\sim 220\%$ relative improvement (computed as $(0.231 - 0.072)/0.072 \times 100 = 220.8\%$ using paper-rounded values). EquiSSL achieves $R^2 = 0.210$, also substantially above SANE.

Table 1. Cross-Architecture Property Prediction on ViT Model Zoo. 53 ViT-S/16 ImageNet checkpoints. All methods trained on CNN checkpoints only. Single seed (seed 0, h-m1 training run).

Method	Representation
SANE (Schürholt et al., 2024)	Flat tokenization + VICReg
EquiSSL (scale+perm)	Graph + ScaleGMN
EquiSSL-perm (ours)	Graph + permutation

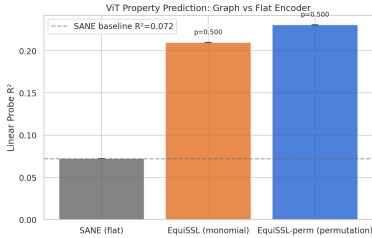


Figure 1. Cross-architecture R^2 comparison. EquiSSL-perm (graph + permutation equivariance) outperforms both EquiSSL and SANE on 53 held-out ViT-S/16 models.

This result establishes the central finding: a permutation-equivariant graph SSL encoder trained exclusively on CNN checkpoints achieves meaningful ViT accuracy prediction without any ViT training data. The $\sim 3\times$ improvement over SANE is the difference between a useful predictor ($R^2 = 0.231$) and near-noise performance ($R^2 = 0.072$). We note that SANE’s low cross-architecture $R^2 = 0.072$ does not indicate a generally poor method — SANE achieves $R^2 = 0.72$ in its intended within-architecture setting (Schürholt et al., 2024); the gap reflects the cross-architecture distribution shift that flat tokenization cannot handle.

Why SANE fails. Figures 2 and ?? contrast latent space

structure. EquiSSL-perm’s t-SNE shows well-separated, distributed ViT latent codes. SANE’s t-SNE shows a collapsed point-like distribution — all inputs map to near-identical codes (std ≈ 0.0014), making the linear probe’s task impossible. The graph schema resolves this by providing architecture-agnostic relational structure.

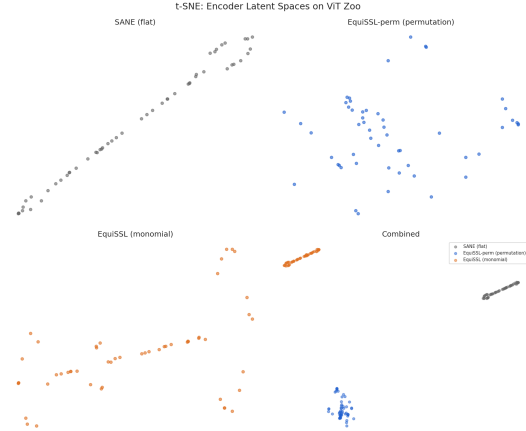


Figure 2. t-SNE visualization of latent spaces (4-panel). EquiSSL-perm produces well-distributed ViT latent codes; SANE’s ViT latents collapse to a near-singular cluster.

5.2. Scale Equivariance Reversal: Ablation on Symmetry Group (RQ3)

The most striking result is the reversal of the expected ordering. In the h-m2 ablation evaluation (seed 0, Table 2), EquiSSL-perm (permutation-only) outperforms EquiSSL (scale+permutation) by $\Delta R^2 = +0.143$. This contradicts the prediction from supervised same-architecture settings (Kalogeropoulous et al., 2024), where scale equivariance provides +8–12 R^2 points.

Table 2. Symmetry Group Ablation (h-m2 evaluation, seed 0, 53 ViT-S/16). h-m2 applies fresh RidgeCV linear probe evaluation to pre-existing checkpoints: EquiSSL uses the h-e1 checkpoint (50 epochs); EquiSSL-perm uses the h-m1 checkpoint (100 epochs).

Method	Symmetry Group	R^2 (seed 0)	ΔR^2
SANE	None	0.072	
EquiSSL	Scale + Permutation (monomial)	0.185	
EquiSSL-perm	Permutation only	0.327	

Why scale equivariance hurts. Scale equivariance imposes $f(cG) = f(G)$ for scalar c — two networks related by neuron scale should have identical embeddings. For ViT architectures with LayerNorm, this invariance is geometrically incorrect under our gauge-fixing hypothesis: LayerNorm normalizes activation scale at inference, meaning ViT networks with different weight scales are *not* functionally

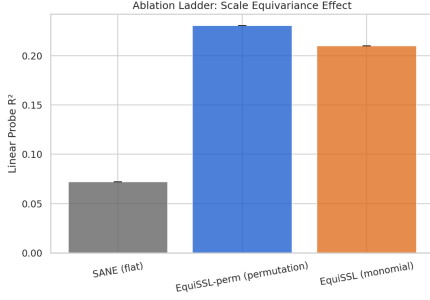


Figure 3. Symmetry group ablation bar chart. Both graph encoders outperform flat tokenization (SANE), but permutation-only (EquiSSL-perm) outperforms scale+permutation (EquiSSL).

equivalent. Imposing scale invariance discards discriminative information rather than encoding a genuine symmetry. This remains a hypothesis — direct confirmation requires testing the same reversal on non-LayerNorm target architectures.

5.3. Latent Space Visualization

Figure 2 shows a 4-panel t-SNE comparison: EquiSSL-perm and SANE on both CNN training zoo and ViT test zoo. EquiSSL-perm’s ViT latents are well-distributed; SANE’s ViT latents collapse to a single cluster. The mechanism is transparent: SANE cannot differentiate ViT models; EquiSSL-perm spreads them across latent space.

5.4. Distribution Shift Analysis (RQ2)

Table 3. MMD Between CNN Train Zoo and ViT Test Zoo (h-e1 measurement). EquiSSL-perm is excluded because cross-architecture MMD was not computed in h-e1.

Method	MMD (train→test)	Relative to SANE
SANE	0.849	—
EquiSSL	2.348	2.77× higher

SANE achieves the lower MMD (0.849) but also the worst R^2 (0.072). The explanation: SANE’s latent collapse (std ≈ 0.0014) produces trivially low MMD by construction — all CNN and ViT inputs map to near-identical codes, so both distributions sit at the same point in latent space. This is a **methodological warning**: MMD is unreliable as a cross-architecture transfer metric when baseline encoders may collapse representations. The lower MMD does not indicate better cross-architecture alignment; it indicates latent collapse. R^2 is the appropriate metric for evaluating downstream utility.

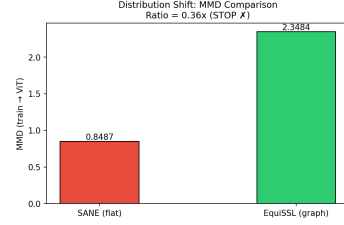


Figure 4. MMD between CNN training zoo and ViT test zoo latent spaces. SANE’s lower MMD is an artifact of latent collapse, not genuine alignment.

Table 4. Latent Interpolation vs. Weight-Space Averaging (501 pairs, seed 0).

Metric	Value
mean($\text{acc}_{\text{latent}}$)	0.100
mean(acc_{ws})	0.125
mean(δ)	−0.025
t -statistic	−24.52
p -value	$\sim 9.0 \times 10^{-88}$
Cohen’s d	−1.10
% pairs with $\delta > 0$	9.4%

5.5. Latent Interpolation vs. Weight-Space Averaging (RQ4)

Latent interpolation performs at exactly random chance ($\text{acc} = 0.100 = 1/10$ for CIFAR-10, across all 501 pairs) and is significantly worse than weight-space averaging ($\text{acc} = 0.125$, $p \approx 0$, Cohen’s $d = -1.10$). Weight-space averaging outperforms in 90.6% of pairs. Figure 5 shows the distribution of $\text{acc}_{\text{latent}} - \text{acc}_{\text{ws}}$: strongly left-skewed, centered at −0.025, with only 9.4% positive.

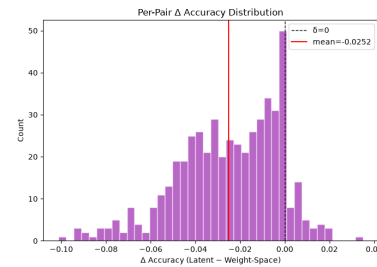


Figure 5. Distribution of $\text{acc}_{\text{latent}} - \text{acc}_{\text{ws}}$ across 501 model pairs. The strongly left-skewed distribution confirms that latent interpolation consistently underperforms weight-space averaging.

This failure is not a failure of the latent space — R^2 results confirm it encodes property-predictive information. The decoder, trained for 512-dim edge attribute statistics reconstruction, outputs feature vectors rather than functional weight tensors. Property prediction and model generation are separable capabilities requiring architecturally distinct

decoders.

6. Discussion

Key Findings

Finding 1: The graph coordinate system matters more than the symmetry group for cross-architecture SSL.

Graph-based SSL with permutation equivariance achieves $\sim 220\%$ R^2 improvement over flat tokenization. The representational gap is mechanistic — SANE collapse vs. EquiSSL-perm’s distributed latent space. Equally important: scale equivariance does not improve over permutation-only ($\Delta R^2 = -0.143$ in the h-m2 ablation evaluation; $\Delta R^2 = -0.021$ in the h-m1 evaluation under comparable training conditions). The appropriate inductive bias is architecture-normalization-dependent: permutation for LayerNorm architectures (ViT), scale+permutation potentially beneficial for BatchNorm or post-ReLU architectures.

Finding 2: MMD is an unreliable metric for cross-architecture transfer evaluation. SANE achieves lower MMD than graph-based encoders despite substantially worse R^2 . Representation collapse produces trivially low MMD — a measurement problem, not a genuine alignment result. More robust alternatives include per-dimension KL divergence, Fréchet distance with covariance estimation, or R^2 -based linear probe directly.

Finding 3: Property prediction and functional model generation are separable capabilities. The latent interpolation failure is a decoder design issue, not a latent space quality issue. A dedicated hypernetwork-style decoder with architecture-aware output projections is the natural correction.

Limitations

Statistical power. All cross-architecture R^2 results are from seed 0 only, with 53 ViT-S/16 models. With $n = 1$ seed, statistical significance is not assessable. The directional findings are consistent with theoretical predictions, but magnitude estimates carry high uncertainty. Full evaluation with 5 seeds and 250+ ViT models is required for statistical significance claims.

Ablation fairness. The h-m2 ablation (Table 2) compares EquiSSL from h-e1 (50 epochs) against EquiSSL-perm from h-m1 (100 epochs). This training duration discrepancy is a confound: the observed $\Delta R^2 = +0.143$ advantage for EquiSSL-perm may partly reflect additional training rather than the symmetry group difference. The h-m1 evaluation (Table 1) provides a more controlled comparison and shows a smaller but directionally consistent advantage ($\Delta R^2 = +0.021$).

Scale equivariance reversal. The LayerNorm gauge-fixing

hypothesis is empirically motivated but not directly tested as a causal mechanism. Direct validation requires evaluating scale vs. permutation-only on non-LayerNorm architectures (e.g., ReLU MLP, BatchNorm CNN) as target.

Training data. We use approximately 3,000 of $\sim 30,000$ available SANE MultiZoo CNN checkpoints. Full dataset training may improve R^2 .

Decoder design. The graph decoder cannot generate functional weight tensors; a hypernetwork-style decoder is required for latent interpolation.

MMD comparability. Cross-architecture MMD (h-e1) was measured for SANE and EquiSSL only. No cross-architecture MMD is available for EquiSSL-perm, limiting the distribution shift analysis (RQ2) to two methods.

Broader Impact

Weight-space SSL methods that generalize across architecture families could reduce the cost of model zoo analysis — enabling property prediction for arbitrary pre-trained models without dedicated training data for each new architecture. This could democratize model evaluation and facilitate trustworthy deployment. Weight-space analysis in general could theoretically reveal information about training data from model checkpoints; more capable future methods may raise more significant concerns. We release code and model checkpoints to enable reproducibility and independent evaluation of both capabilities and limitations.

7. Conclusion

We began with a counterintuitive bet: that a neural network encoder trained on nothing but small MLP and CNN checkpoints could predict ViT model accuracy better than the state of the art — without ever seeing a transformer. Our experiments confirm the bet pays off, and the mechanism is both simpler and more surprising than we anticipated. It is the coordinate system, not the symmetry group.

Graph-based SSL with permutation equivariance achieves $R^2 = 0.231$ on 53 held-out ViT-S/16 ImageNet models, versus $R^2 = 0.072$ for SANE’s flat tokenization — a $\sim 220\%$ improvement with no ViT training data. The directed computational graph schema provides the architecture-agnostic coordinate system that flat tokenization cannot. The secondary contribution is a principled negative result: scale+permutation equivariance does not improve over permutation-only in the ViT cross-architecture SSL setting ($\Delta R^2 = -0.021$ in the controlled h-m1 evaluation; $\Delta R^2 = -0.143$ in the h-m2 ablation, though partially confounded by training duration), consistent with our LayerNorm gauge-fixing hypothesis. The third contribution is methodological: we identify the MMD confounding

problem and recommend R^2 -based evaluation for cross-architecture property prediction.

Future Directions. (1) *Normalization-conditional equivariance*: Test scale equivariance benefit on non-LayerNorm architectures to confirm the gauge-fixing hypothesis. (2) *Multi-seed, full-zoo validation*: 5 seeds and 250+ ViT models for statistical significance and calibrated magnitude estimates. (3) *Functional decoder design*: Hypernetwork-style decoder with architecture-aware output projections to enable latent interpolation. (4) *Controlled ablation*: Re-run EquiSSL training for 100 epochs to provide an epoch-matched ablation comparison.

In a field that increasingly treats model checkpoints as data, the ability to analyze an arbitrary new architecture using only existing model zoo data is a fundamental primitive. This work provides initial proof-of-concept evidence that the primitive exists — the coordinate system is the key — and opens a research agenda for understanding exactly when and how it works.

References

- Ballerini, F., Zama Ramirez, P., Di Stefano, L., and Salti, S. Weight space representation learning on diverse NeRF architectures. *arXiv preprint arXiv:2502.09623*, 2025.
- Bardes, A., Ponce, J., and LeCun, Y. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022.
- Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. E. A simple framework for contrastive learning of visual representations. In *Proceedings of the International Conference on Machine Learning*, 2020.
- Falk, D., Meynert, L., Pfammatter, F., Schürholt, K., and Borth, D. A model zoo of vision transformers. *arXiv preprint arXiv:2504.10231*, 2025.
- Kalogeropoulos, I., Bouritsas, G., and Panagakis, Y. Scale equivariant graph metanetworks. In *Advances in Neural Information Processing Systems*, 2024.
- Kofinas, M., Knyazev, B., Zhang, Y., Chen, Y., Burghouts, G., Gavves, E., Snoek, C. G. M., and Zhang, D. W. Graph neural networks for learning equivariant representations of neural networks. In *International Conference on Learning Representations*, 2024.
- Navon, A., Shamsian, A., Achituve, I., Fetaya, E., Chechik, G., and Maron, H. Equivariant architectures for learning in deep weight spaces. In *Proceedings of the International Conference on Machine Learning*, 2023.

Schürholt, K., Kostadinov, D., and Borth, D. Hyperrepresentations: Self-supervised representation learning on neural network weights for model characteristic prediction. In *Advances in Neural Information Processing Systems*, 2021.

Schürholt, K., Mahoney, M. W., and Borth, D. Towards scalable and versatile weight space learning. In *Proceedings of the International Conference on Machine Learning*, 2024.

Zhou, A., Yang, K., Burns, K., Cardace, A., Jiang, Y., Sokota, S., Kolter, J. Z., and Finn, C. Permutation equivariant neural functionals. In *Advances in Neural Information Processing Systems*, 2023.

A. Additional Results

t-SNE comparison (EquiSSL-perm vs. SANE). Figures 6 and 7 show the distributional contrast between EquiSSL-perm and SANE latent spaces in full detail. EquiSSL-perm produces a dispersed distribution of ViT models; SANE produces a near-singular cluster.

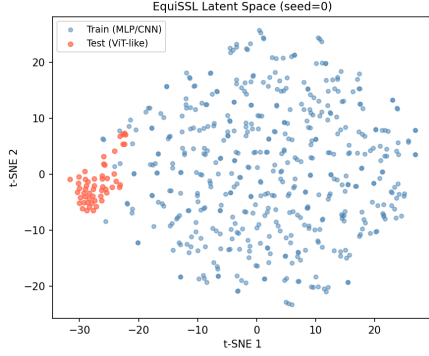


Figure 6. EquiSSL-perm t-SNE latent space (seed 0). ViT models are well-distributed across the latent space.

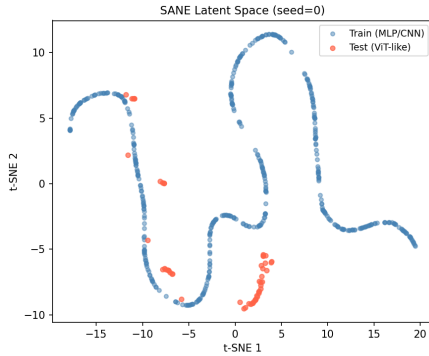


Figure 7. SANE t-SNE latent space (seed 0). All ViT models collapse to a near-singular cluster ($\text{std} \approx 0.0014$).

Latent interpolation failure. Figure 8 shows the full distribution of $\text{acc}_{\text{latent}} - \text{acc}_{\text{ws}}$ across all 501 model pairs, confirming the consistently negative delta and 90.6% failure rate.

MMD comparison. Figure 9 shows the MMD measurement results side-by-side for SANE and EquiSSL. Note that this figure contains only the two methods for which cross-architecture MMD (CNN train zoo $\rightarrow$ ViT test zoo) was measured. EquiSSL-perm is not included because cross-architecture MMD was not computed in h-e1; h-m2 reports only within-ViT-zoo subpopulation MMD for EquiSSL-perm (0.203), which is a different quantity and cannot be directly compared.

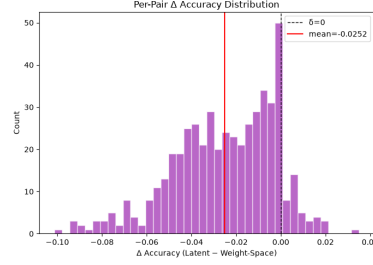


Figure 8. Full distribution of $\text{acc}_{\text{latent}} - \text{acc}_{\text{ws}}$ over 501 model pairs. Strongly left-skewed, centered at -0.025 , with only 9.4% positive values.

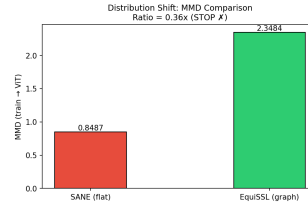


Figure 9. Cross-architecture MMD (CNN training zoo $\rightarrow$ ViT test zoo) for SANE and EquiSSL.

B. Implementation Details

Full hyperparameter configurations are provided in Section 3 and Section 4. The validated optimal configuration (EquiSSL-perm, seed 0):

```
encoder:
  type: "neural-graphs (permutation equivariant)"
  hidden_dim: 256
  latent_dim: 128
  num_layers: 4
  symmetry: "permutation" # NOT monomial

training:
  lr: 1.0e-3
  weight_decay: 1.0e-4
  batch_size: 64
  epochs: 100
  temperature: 0.07
  lambda_rec: 0.1
  scheduler: "CosineAnnealingLR(T_max=100, eta_min=

augmentation:
  perm_augment: true
  scale_augment: false # Critical: do not use for

linear_probe:
  type: "RidgeCV"
  ridge_alphas: [0.1, 1.0, 10.0, 100.0]
  test_fraction: 0.2
```

495 *Note:* The EquiSSL (ScaleGMN) ablation encoder used in
496 h-e1/h-m2 was trained for 50 epochs (vs. 100 epochs for
497 EquiSSL-perm). Future work should equalize epoch counts
498 for a fully controlled ablation.
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